A PROJECT TITLE

ON

Diabetes Risk Prediction Using Logistic Regression and Random Forest with Cross-Validation

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF : Artificial Intelligence and Data Science

Name of Title: Diabetes Risk Prediction Using Logistic Regression and Random Forest with Cross-Validation

1. Abstract:

Diabetes is one of the most common chronic diseases worldwide, affecting millions of people and leading to serious health complications if not diagnosed early. Early prediction of diabetes helps healthcare professionals provide timely treatment, improve patient outcomes, and reduce medical costs. This project presents a machine learning-based approach for predicting diabetes risk using **Logistic Regression** and **Random Forest** algorithms. The objective is to compare the performance of these two classification models and identify the most effective method for accurate diabetes prediction.

The project uses a publicly available diabetes dataset containing medical attributes such as glucose level, blood pressure, body mass index (BMI), insulin level, age, skin thickness, pregnancies, and diabetes pedigree function. Before training the models, the dataset undergoes preprocessing steps, including handling missing values, feature scaling, and data normalization where required. The data is then divided into training and testing sets.

To improve the reliability and generalization of the models, **K-Fold Cross-Validation** is applied during the training process. Logistic Regression serves as a simple and interpretable baseline classifier, while Random Forest, an ensemble learning technique, captures complex relationships among features and improves prediction accuracy. The performance of both models is evaluated using metrics such as **Accuracy, Precision, Recall, F1-Score, and ROC-AUC Score**.

The experimental results are expected to demonstrate that Random Forest achieves higher prediction accuracy than Logistic Regression due to its ability to reduce overfitting and handle nonlinear relationships effectively. The developed system can assist healthcare professionals in identifying individuals at high risk of diabetes and support early preventive measures. This project highlights the importance of machine learning in healthcare and demonstrates how predictive analytics can contribute to faster, more accurate, and cost-effective medical decision-making.

2. INTRODUCTION:

Diabetes is a chronic metabolic disorder that occurs when the body cannot effectively regulate blood sugar levels. Delayed diagnosis may result in severe health complications such as heart disease, kidney failure, nerve damage, and vision loss. Traditional diagnosis often depends on laboratory tests and clinical expertise. Machine Learning provides an efficient solution by analyzing patient medical data and predicting the likelihood of diabetes at an early stage.

This project develops a diabetes prediction model using **Logistic Regression** and **Random Forest** algorithms with **K-Fold Cross-Validation**. The objective is to compare both algorithms and determine which model provides better predictive performance. The proposed system aims to assist doctors and healthcare organizations in making faster and more accurate decisions while reducing diagnostic time and improving patient care.

3. SOFTWARE REQUIREMENTS:

Operating System Windows: 10/11

Programming Language : Python 3.10

IDE : Jupyter Notebook / Visual Studio Code / PyCharm

Dataset : PIMA Indians Diabetes Dataset (Kaggle/UCI Repository)

Libraries : NumPy, Pandas, Matplotlib, Seaborn

Machine Learning Library : Scikit-learn

Model Evaluation : Accuracy Score, Precision, Recall, F1-Score, ROC-AUC, Cross-Validation

Version Control: Git & GitHub

Signature of the Guide Course Instructor